

Chapter 12



Purchasing & Supply Chain Analysis: Tools & Techniques

Strategic Sourcing & Procurement in Supply Chain Management

KFUPM Business School | Masters Program

Opening Case: Aramco's Supplier Development Initiative

Setting the Stage — Real-World Context

CASE EXAMPLE

The Challenge

Saudi Aramco — one of the world's largest oil & gas companies — needed to qualify 40 new domestic suppliers within 18 months to comply with In-Kingdom Total Value Add (IKTVA) targets.

The Procurement Team's Toolkit

- Used Project Management (PERT/CPM) to schedule 40 parallel supplier audits
- Applied Learning Curve analysis to forecast supplier cost reductions over time
- Ran Value Analysis sessions to strip non-essential specs from 12 product categories
- Used Process Mapping to reduce qualification cycle from 9 months → 5 months



Saudi Aramco
Dhahran, Eastern Province
Kingdom of Saudi Arabia

44%

reduction in qualification time



Opening Discussion

Before we dive in — think about this:

"A major Saudi company asks you to reduce procurement costs by 20% next year — without switching suppliers. Which analytical tools would you reach for first, and why?"

 **Share your thoughts with a partner — 2 minutes**

Today's Lecture Roadmap

Five Core Analytical Tools in Strategic Sourcing



01 Project Management



02 Learning Curve Analysis



03 Value Analysis / VE



04 Quantity Discount Analysis



05 Process Mapping

Tool 1: Project Management

Organizing Complex Sourcing Initiatives

CONCEPT

What Is a Project?

- ✓ Specific objectives with clear deliverables
- ✓ Defined start and end dates
- ✓ Consumes limited resources (time, people, budget)
- ✓ Cross-functional team involvement

Supply Chain Examples:

- New product development
- Sourcing strategy rollout
- MIS implementation
- Supplier development

Project Success Criteria

On time & on budget

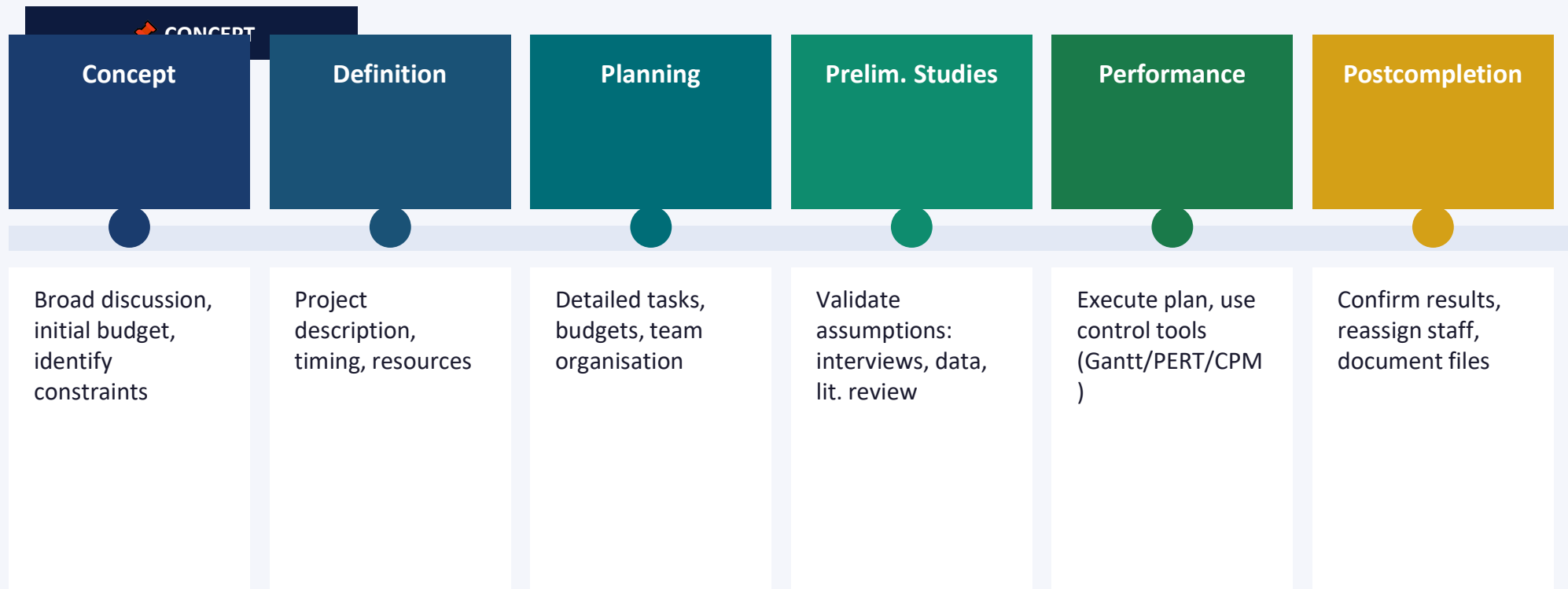
Meets specifications

Satisfies stakeholders

Minimal scope changes

Doesn't disrupt operations

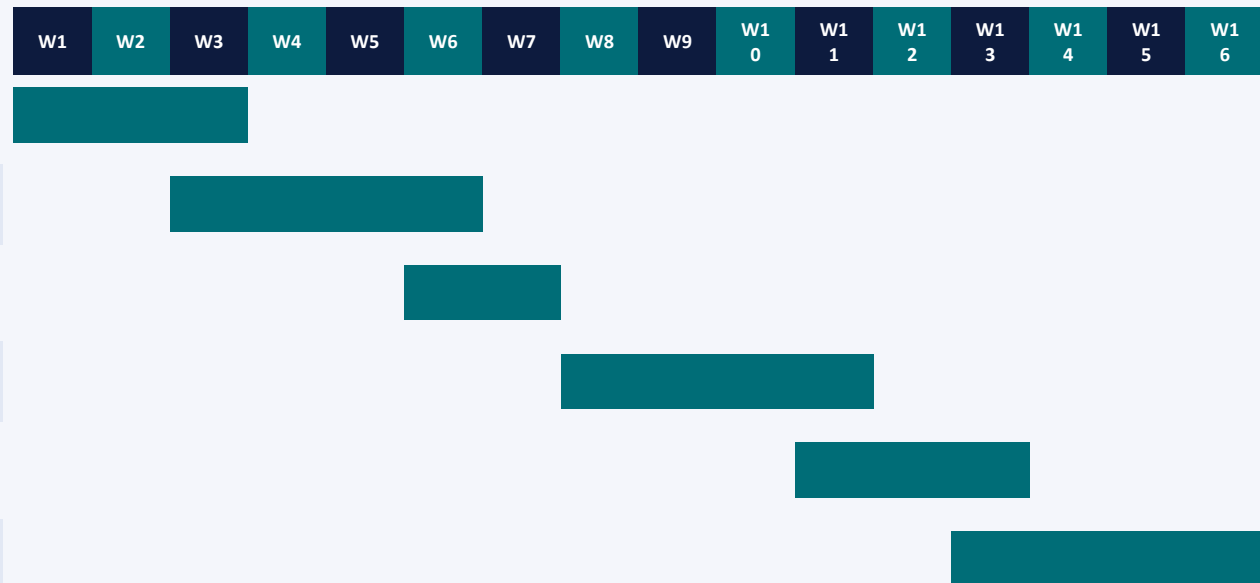
Project Phases: From Concept to Completion



Project Planning Tool: Gantt Chart

Visualising Tasks, Timing & Dependencies

 EXAMPLE



 Gantt charts show what runs in parallel vs. sequentially — critical for multi-supplier projects.

CPM vs. PERT: Choosing the Right Tool

CPM

Critical Path Method

Use when:

- Single known time per activity
- Low uncertainty
- Deterministic environment

Example: Equipment installation schedule at a fully contracted supplier plant



PERT

Program Eval & Review Technique

Use when:

- Time estimates are uncertain
- Uses 3 estimates: Optimistic / Most Likely / Pessimistic

Example: New supplier onboarding where audit duration varies by country

Simulation Practice: PERT Calculation

Hands-On Application



Scenario: You are managing a supplier qualification project at a SABIC plant. Estimate expected duration for each activity.

Activity	Optimistic (O)	Most Likely (M)	Pessimistic (P)	Expected Te = ?
Supplier site audit	3 days	5 days	10 days	_____ days
Contract negotiation	2 days	4 days	9 days	_____ days
Quality certification	5 days	8 days	14 days	_____ days
System integration test	1 day	3 days	7 days	_____ days

PERT Formula

$$Te = (O + 4M + P) / 6$$

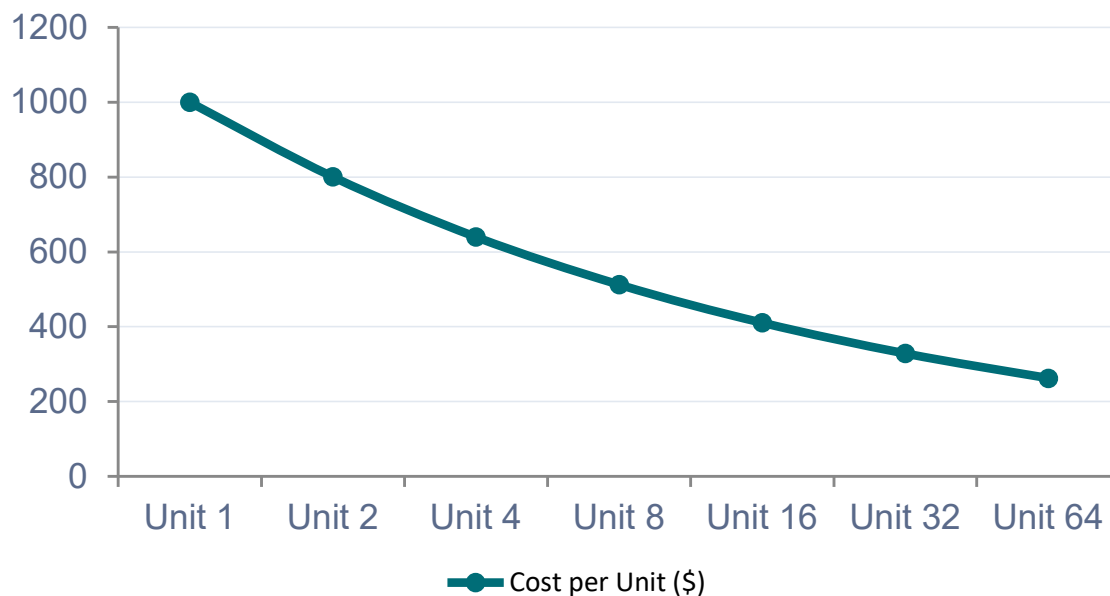
O=Optimistic M=Most Likely P=Pessimistic

Calculate Te for each row, then identify which activity has the highest uncertainty (P – O). 10 minutes.

Tool 2: Learning Curve Analysis

Forecasting Cost Reductions as Production Scales

80% Learning Curve (Cost Drops as Output Doubles)



Key Insight

Each time cumulative production DOUBLES, the cost per unit falls to a fixed percentage of its previous level.

80% curve: cost drops 20% every doubling

When to use it:

- New process at supplier
- Complex, labour-intensive items
- Negotiating long-term contracts

Real Case: Boeing & the 80% Learning Curve

How Aerospace Procurement Uses Learning Curves

Boeing applies learning curve analysis when negotiating with suppliers for aircraft components — it is industry standard.

The Context

Boeing's 737 fuselage sections are supplied by Spirit AeroSystems. As Spirit ramps up production, Boeing's procurement team expects predictable cost reductions.

The Tool Applied

Boeing negotiators use an 85% learning curve model. If Unit 1 costs \$2M, Unit 2 should cost ~\$1.7M, Unit 4 ~\$1.45M, and so on.

The Negotiation Edge

Buyers use learning curve projections to reject price quotes that don't reflect expected efficiency gains — saving billions across multi-year contracts.

KFUPM Relevance

Saudi defence and energy companies (GAMI, Aramco) increasingly apply this in domestic procurement to drive IKTV supplier efficiency.

Tool 3: Value Analysis / Value Engineering

Maximising Function, Minimising Cost

$$\text{VALUE} = \text{FUNCTION} \div \text{COST}$$

1. Gather Info

What does the product do? Primary vs. secondary functions. Name each with noun + verb.

2. Speculate

Brainstorm: generate every improvement idea without judgment.

3. Analyse

Cost/benefit analysis. Feasibility assessment of all ideas.

4. Recommend

Present to management. Requires analytical skill + communication.

5. Follow Up

Ensure progress. Document outcomes for continuous improvement.

Real Case: Toyota's Value Engineering Culture

VA/VE as a Competitive Advantage

Toyota Production System & VA/VE

Background:

Toyota's procurement philosophy mandates that every purchased component undergoes Value Engineering before production freeze. Suppliers and buyers jointly review designs.

How VA Questions Were Applied:

- "Can a standard part replace this custom bracket?" → saved \$4 per unit
- "Are specs too tight for actual customer needs?" → relaxed tolerances cut machining cost 18%
- "Can we combine two parts into one?" → reduced assembly time and inventory

VA Test Questions

1. Does it contribute value to customers?
2. Is cost proportionate to usefulness?
3. Can weight be reduced?
4. Can a standard part replace custom?
5. Can we buy it cheaper elsewhere?
6. Are specs tighter than needed?



Group Discussion Activity

Value Analysis in Practice

Scenario:

A Saudi packaging company supplies custom plastic containers to a major FMCG brand. Current unit cost is SAR 12. The procurement manager wants to apply VA/VE to reduce cost by 15% without affecting product functionality.

Your Task (Groups of 3–4):

1. List 3 VA questions you would ask the supplier
2. Identify 2 specific cost-reduction ideas that preserve function
3. Present your best idea to the class (2 min per group)

 15 minutes | All groups present

Tool 4: Quantity Discount Analysis

Are the Discounts Really Worth It?

★ CONCEPT

A buyer must verify that quantity discounts are genuinely justified by the supplier's incremental cost structure — not arbitrary.

Units / Order	1	3	6	10
Price / Unit (Quoted)	\$85.00	\$80.00	\$70.00	\$69.00
Total Price / Order	\$85.00	\$240.00	\$420.00	\$690.00
Price Difference	\$85.00	\$155.00	\$180.00	\$270.00
Qty Difference	1	2	3	4
Incremental Cost / Unit ⚠	\$85.00	\$77.50	\$60.00	\$67.50 ↑

🔍 **Key Insight:** The jump from 6 units (\$60/unit) to 10 units (\$67.50/unit) means the discount is LESS at higher quantity — a red flag for negotiation!

Simulation: Negotiating with Quantity Discount Data



SIMULATION

Role Play: You are a procurement manager at a Saudi petrochemical company purchasing industrial valves. The supplier has provided the following pricing:

Order Qty	50 units	100 units	200 units	500 units
Price/Unit	SAR 480	SAR 450	SAR 400	SAR 390
Total Cost	SAR 24,000	SAR 45,000	SAR 80,000	SAR 195,000
Incremental Cost/Unit	SAR 480	SAR 420	SAR 350	SAR ?

Discussion Questions:

1. Calculate the incremental cost for the 500-unit order.
2. Is the 500-unit discount justified? What would you say to the supplier?

Tool 5: Process Mapping

Eliminating Waste in Supply Chain Processes

What is Process Mapping?

Breaks a process down to individual steps to identify:

 **Value-Adding Activities:** Transform materials toward customer value

 **Non-Value-Adding Activities:** Waste — delays, duplication, unnecessary inspection

Types:

- Sequential Processes
- Concurrent Processes

Best used with cross-functional teams — generates buy-in from all affected stakeholders.

BEFORE

215 minutes
to unload one trailer
(7 steps, 4 non-value-adding)

AFTER MAPPING

Eliminated steps 4 & 5
Combined steps 2 & 3
~30% time reduction possible

Process Mapping Example: Warehouse Receiving

Identifying Value-Adding vs. Non-Value-Adding Steps

Step	Activity	Time	VA?
1	Place trailer at receiving dock	15 min	✓ VA
2	Unload trailer with equipment	30 min	✓ VA
3	Check quantity vs. shipping docs	30 min	✗ NVA
4	Acknowledge receipt & print control tickets	60 min	✗ NVA
5	Attach control tickets to loads	20 min	✗ NVA
6	Inspect inbound material	30 min	✓ VA
7	Move material to warehouse	30 min	✓ VA
Total: 215 minutes NVA Steps: 3, 4, 5 Target: Eliminate or Automate NVA Steps			

Integrative Case Study

SABIC: Qualifying a New Resin Supplier in 90 Days



Background:

SABIC's procurement team must qualify a new polypropylene resin supplier from Malaysia within 90 days to avoid a production shutdown. Budget: SAR 500,000.

Project Mgmt (PERT)

Map all qualification activities. Estimate timeline. Identify the critical path. Where is your biggest risk?

Learning Curve

Supplier quotes SR 4,000/tonne for first order. Expect 85% learning curve. What price should you target by order #4?

Value Analysis

Current spec requires Grade A purity. Run 5 VA questions. Could Grade B work for 70% of products?

Quantity Discount

Supplier offers 5% off at 50T, 12% at 100T, 13% at 200T. Analyse incremental costs. Is 200T justified?

Process Mapping

Current inbound quality inspection takes 8 days. Map the steps. Identify NVA activities. Target: 4 days.

Which Tool — When?

A Quick Decision Guide for Procurement Professionals

SYNTHESIS

 Launching a major supplier development programme	Project Management (Gantt/PERT/CPM)
 Negotiating prices for a complex, first-time manufactured part	Learning Curve Analysis
 An item costs more than comparable alternatives	Value Analysis / Value Engineering
 Supplier offers tiered pricing by order quantity	Quantity Discount Analysis
 Cycle time or lead time seems unnecessarily long	Process Mapping

Exit Ticket — Before You Leave

Consolidate Your Learning

EXIT ACTION

1. Name ONE tool

...you would use to manage a 6-month supplier qualification project.

2. Explain in one sentence

...what an 85% learning curve means for a procurement negotiator.

3. Identify ONE non-value-adding step

...in a process you experience in daily life or work.

4. Pose ONE question

...you still have after today's lecture.

 Write your answers on a notecard and hand it in — or post to the class discussion board.

Key Takeaways

- 1 Procurement professionals need a toolkit — not just intuition — to justify decisions.
- 2 Project Management tools (Gantt, CPM, PERT) make complex sourcing initiatives manageable.
- 3 Learning Curves give buyers quantitative leverage in long-term supplier negotiations.
- 4 Value Analysis asks the fundamental question: 'Is every feature worth its cost?'
- 5 Quantity Discount Analysis prevents overpaying for phantom savings at higher volumes.
- 6 Process Mapping exposes hidden waste and accelerates supply chain cycles.

Next Steps & Assignments

Applying the Tools Beyond the Classroom

ACTION ITEMS

Assignment (Individual)

Select a real Saudi company. Identify ONE procurement challenge they face. Propose which analytical tool from Chapter 12 would best address it and why. (1–2 pages, due next class)

Recommended Reading

Monczka et al., Chapters 12–13. Supplement: ISM Supply Management Handbook, Learning Curve Analysis chapter.

Team Project Milestone

Complete your PERT/CPM network for the supplier qualification case. Identify the critical path and calculate expected project duration. Present Week 10.

Think About This

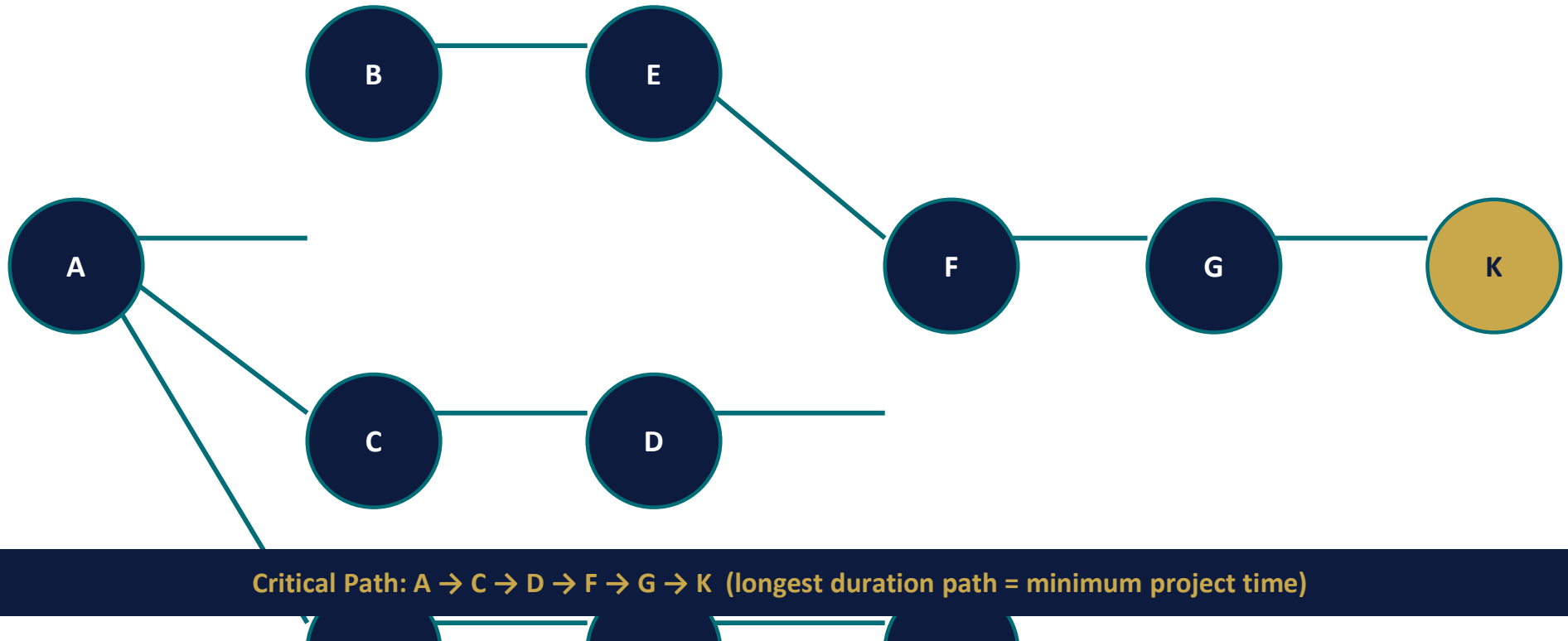
Can you map a 'process' in your personal life? Try it — map your morning routine. Which steps are value-adding?

CPM Network: Supplier Selection Example

Reading & Building Activity Networks

EXAMPLE

Each arrow = one activity. Each circle = an event (milestone). An activity cannot begin until ALL preceding activities are complete.



Learning Curve: Worked Numerical Example

Step-by-Step Cost Projection

WORKED EXAMPLE

Scenario: A new Saudi defence contractor produces armoured vehicle components for the first time. Unit 1 costs SAR 800,000 at an 80% learning rate.

Cumulative Unit	Learning %	Unit Cost (SAR)	Cumulative Savings
1	100%	800,000	—
2	80%	640,000	160,000
4	80%	512,000	448,000
8	80%	409,600	948,800
16	80%	327,680	SAR 1.73M saved

$$\text{Cost}(N) = \text{Cost}(1) \times N^{(\log r / \log 2)}$$

r = learning rate (0.80)
N = cumulative unit number

 **Procurement Insight:** By Unit 16, cost has dropped 59% from Unit 1. Buyers who don't factor this in leave millions on the table.

Learning Curve: When It Does NOT Apply

Critical Thinking for Procurement Professionals



APPLIES WHEN...

- New, complex manufacturing process
- High direct labour content (>30% of cost)
- First production run of a technical item
- Long-term contract with volume growth
- Supplier with demonstrated improvement culture



DOES NOT APPLY...

- Commodity raw materials (steel, chemicals)
- Highly automated processes — robots don't 'learn'
- Short production runs or one-off orders
- Items driven by material cost, not labour
- Markets with volatile input prices

 Discussion: Can you think of a Saudi industry where learning curves are most and least relevant?

VA/VE: Cross-Functional Team Composition

Why Multiple Perspectives Are Essential

Executive Management

Champions resources, approves changes, sets strategic direction

Design Engineering

Spec requirements, technical feasibility of alternatives

Quality Control

Standards compliance, customer requirements, testing data

Suppliers

Material alternatives, process knowledge, cost transparency

Supply Management

Cost data, supplier alternatives, market intelligence

Production / Ops

Manufacturability, process impact, capacity constraints

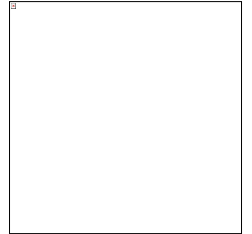
Marketing

Customer value perception, competitive positioning, branding

Finance / Costing

TCO analysis, ROI of changes, budget allocation

Value Engineering in Action



The Product: Industrial Packaging Box (used for exporting petrochemical products)

Current cost: SAR 28/unit | Annual volume: 200,000 units | Total spend: SAR 5.6 million/year

Component	Unit Cost	Function	VA Question to Ask
Corrugated outer shell	SAR 12	Structural protection	_____
Inner foam lining	SAR 8	Vibration dampening	_____
Custom printed label	SAR 4	Brand identification	_____
Metal corner brackets	SAR 3	Edge reinforcement	_____
Moisture-seal tape	SAR 1	Moisture barrier	_____



Groups: Fill in the VA question for each component. Which component offers the highest saving potential? 12 minutes.

Quantity Discounts & the EOQ Trade-off

Balancing Purchase Price vs. Holding Cost

CONCEPT

A lower unit price from bulk buying is only beneficial if the savings outweigh the increased inventory holding costs.

Benefits of Larger Orders

- ✓ Lower unit purchase price
- ✓ Fewer purchase orders → lower ordering cost
- ✓ Less risk of stockouts
- ✓ Stronger supplier relationship leverage

Hidden Costs of Larger Orders

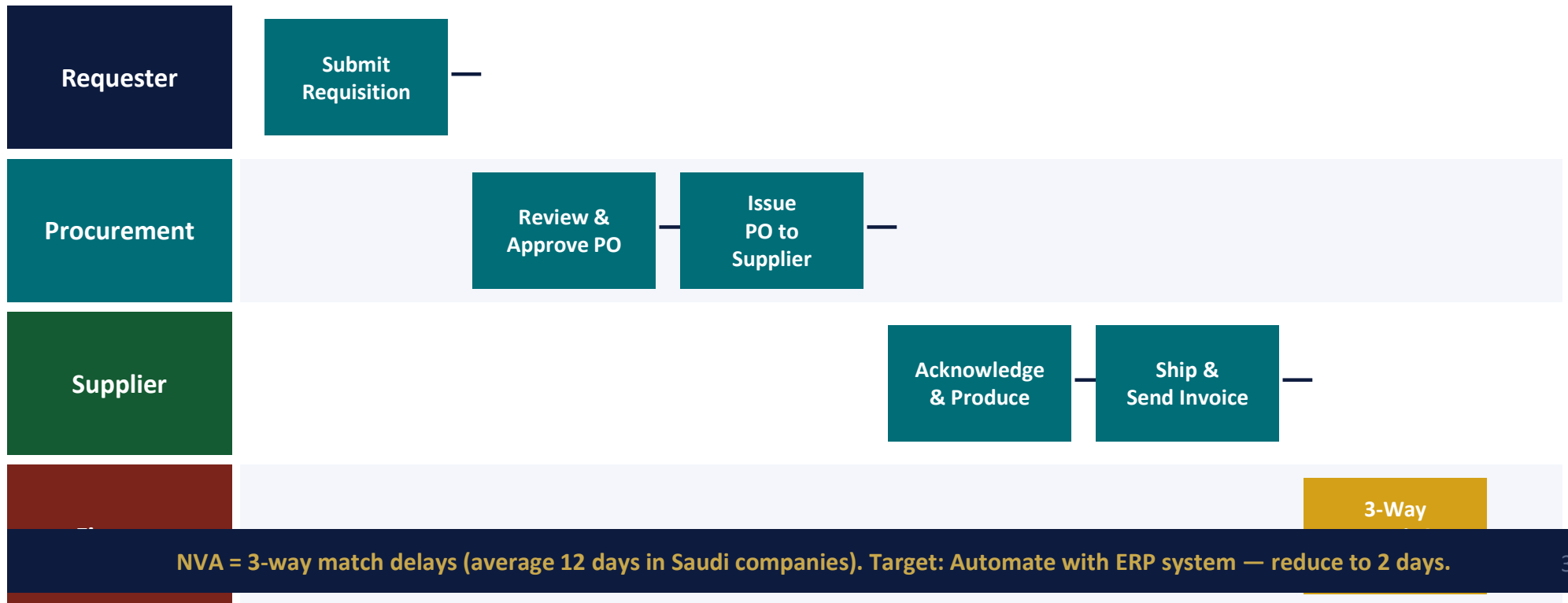
- ✗ Higher inventory holding costs (25–30% of value/year)
- ✗ Greater obsolescence/spoilage risk
- ✗ More warehouse space required
- ✗ Ties up working capital

Process Mapping: Swim-Lane Diagram

Mapping Across Functions to Find Gaps & Delays

 EXAMPLE

Purchase Order Process — from Requisition to Payment (current state)



From Supplier Selection to Contract Award

The Challenge:

Your company's supplier selection process currently takes 47 days from RFQ to contract signing. Leadership has set a target of 25 days.

Step 1	Map all current steps from RFQ issue to contract award. List each activity, who owns it, and average time.	4 min
Step 2	Classify each step: Value-Adding (VA) or Non-Value-Adding (NVA). Use a green/red marker system.	3 min
Step 3	Calculate total VA time vs. NVA time. What % of the 47 days is actually productive?	2 min
Step 4	Propose a redesigned process. Which NVA steps can be eliminated, combined, or automated?	4 min



Best redesigned process wins — present your swim-lane to the class.

Integrating the Five Tools: A Strategic Framework

From Analysis to Competitive Advantage



These tools work together — the output of one often feeds the input of another. Master the combination, not just the individual tools.

Saudi Vision 2030 & These Tools

Local Content, IKTVA, and Strategic Sourcing

Vision 2030 mandates that Saudi companies increase domestic sourcing and develop local suppliers — all five tools in this chapter are directly relevant.

Project Management

Managing IKTVA supplier qualification projects across 40+ categories simultaneously requires structured PM with Gantt & CPM.

Learning Curve Analysis

As new Saudi defence (GAMI) and energy suppliers ramp up, buyers must model and negotiate cost reductions using learning curves.

Value Analysis / VE

Localising imported parts requires VA/VE to determine if domestic specs can achieve same function at competitive cost.

Quantity Discount Analysis

Bulk orders from domestic suppliers need rigorous discount analysis to avoid subsidising inefficiency.

Process Mapping

Procurement processes in government-linked companies often have excessive NVA steps — mapping drives Lean transformation.

Common Mistakes in Practice

What Goes Wrong — And How to Avoid It

▲ CRITICAL THINKING
Project Management
Learning Curve
Value Analysis
Quantity Discount
Process Mapping

✗ Setting unrealistic deadlines without a PERT/CPM analysis — ignoring the critical path
✓ Always identify the critical path first; it defines the minimum possible project duration.

✗ Applying learning curves to fully automated processes where no labour learning occurs
✓ Verify labour intensity before applying. Use only for high direct-labour content items.

✗ Removing features without customer/user validation — reducing function alongside cost
✓ Always test changes against the primary function. Cost savings that destroy value are counterproductive.

✗ Accepting a discount without analysing incremental unit cost — falling for 'phantom savings'
✓ Calculate incremental cost per additional unit. A declining discount rate is a red flag.

✗ Mapping the process as it should work, not as it actually works — hiding the real waste
✓ Walk the actual process ('go to gemba'). Interview the people doing the work, not their managers.

How have you seen this mistake made in your organisation?

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The Procurement Analyst Challenge

Final Closing Activity — Put It All Together

"You are a procurement analyst at a KFUPM spin-off technology company. In one week, you must present a cost reduction plan worth SAR 2 million to the CFO."

Which 3 tools from today's lecture would you use, in what order, and why?"



Individual Reflection — 2 minutes. Then share with the class